

[REFERENCE IMPLEMENTATION]

CAUSAL INFERENCE · REVENUE OPERATIONS · PORTFOLIO

Causal Uplift Renewal-Discount Targeting

A heterogeneous-treatment-effect model that targets renewal discounts by who a discount actually persuades — not by who is most likely to churn.

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T-learner uplift model · Qini validation · discount-spend backtesting



Spend the discount budget where it changes the renewal decision

This project builds a causal uplift model that estimates each renewing account's individual treatment effect (ITE) from a retention discount — the difference between renewal probability with and without the offer. Rather than discounting everyone or targeting whoever looks likely to churn, it targets the accounts a discount actually swings.



55%

Discount-spend reduction versus blanket discounting, same targeting logic



77.5%

Precision of the top-45% targeted segment on true persuadable accounts



0.522

Correlation between predicted uplift and simulated ground-truth effect



30,000

Renewal accounts — 21,000 train / 9,000 held-out test, true randomization



Read honestly

The Qini coefficient (0.080) sits below the briefing's 0.15 aspirational target, which assumes a causal-forest upgrade (Slide 30). The spend and precision wins are real; the uplift-ranking quality has a clear, disclosed next step.

Presentation agenda



01

The Business Problem

Why blanket discounting and churn scores both fail



02

Data & Experiment Design

30,000 accounts under a real randomized holdout



03

Feature Engineering & Segments

Ten features across firmographics and engagement



04

Modeling Approach — T-Learner

Two response surfaces, one individual treatment effect



05

Evaluation & Validation

Qini curves, ground-truth correlation, honest gaps



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Business Impact & Targeting

Spend reduction and precision on the persuadable segment



07

Deployment & Responsible Use

Dashboard, model card, experiment-design guardrails

[01 / 07]

SECTION ONE

The Business Problem

Two conventional playbooks for renewal discounting both leave money on the table — one overspends, the other aims at the wrong target.

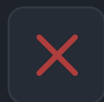
SECTION 1 OF 7



SEC. 01

Blanket discounting spends against every renewal

The simplest retention play is to offer every renewing account a discount. It works — renewal rates are higher with an offer than without — but it also pays full discount cost on accounts that would have renewed anyway.



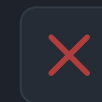
Would renew anyway

A discount here is pure margin loss — the account never needed persuading.



Genuinely persuadable

A discount here is the entire point — it changes a churn into a renewal.



Would churn regardless

A discount here is also wasted — no offer at this price point changes the outcome.

Only the middle group — the persuadable segment — justifies the discount's cost. Blanket discounting pays for all three groups at once; this project builds the model that tells you which group each account is in.

Churn-propensity scores target the wrong axis

The natural next step is to discount accounts with the highest predicted churn risk. But churn risk and discount responsiveness are different questions — and conflating them routinely misdirects the budget.

■ High churn risk, high uplift

The ideal target — at risk, and a discount changes the outcome.

■ High churn risk, low uplift

“Lost causes” — a propensity model chases these hardest; a discount rarely saves them.

■ Low churn risk, high uplift

Easy to miss with propensity scoring — not obviously at risk, but genuinely persuadable.

■ Low churn risk, low uplift

Would renew regardless — harmless to skip, wasteful to discount.

Predict the effect of the offer, not the outcome

A churn model predicts $P(\text{renew})$. An uplift model predicts a difference: $P(\text{renew} \mid \text{offered discount}) - P(\text{renew} \mid \text{no discount})$ for the same account — its individual treatment effect (ITE). That difference, not the raw renewal probability, is what should drive targeting.

The ITE formula

$$\text{ITE}(x) = P(\text{renew} \mid X=x, \text{treated}) - P(\text{renew} \mid X=x, \text{control})$$

Estimated per account from two separately trained response models — not observed directly, since no single account is ever both treated and untreated (Slide 21).

Why this reframing matters



Ranks accounts by responsiveness to the specific lever (a discount), not general risk



Naturally excludes both "sure things" and "lost causes" from the target list



Requires randomized treatment data to estimate validly — sets up Section 02

What this project set out to prove



Estimate individual treatment effect

Score every renewing account with a per-account causal uplift estimate, not a propensity score.



Validate against ground truth

Use the synthetic cohort's known true effect to directly check uplift-recovery quality — impossible in most real deployments.



Cut discount spend, not renewals

Show a targeting rule that spends less while protecting the renewal rate the discount programme exists to defend.



Report the Qini gap honestly

Benchmark against the standard uplift metric and disclose exactly where this model falls short of the aspirational target.



Respect the experiment design

Treat the randomized holdout as a hard requirement, not a formality — and say so for future deployments.



Package for stakeholder review

Deliver a model card, Qini report, and dashboard a RevOps leader can actually act on.

[02 / 07]

SECTION TWO

Data & Experiment Design

30,000 renewal accounts under a true randomized holdout — the one precondition that makes causal targeting valid.

SECTION 2 OF 7



SEC . 02

A randomized renewal-account cohort

30,000 renewal accounts, each with firmographic and engagement features, a randomized discount-offer flag, and an observed renewal outcome. A synthetic ground-truth causal effect is also retained — for validation only, never for training.

Schema (14 columns × 30,000 rows)

account_id	Unique identifier	Key
arr, seat_count, contract_length_months	Firmographic account size fields	Feature
tenure_months, prior_discount_count	Relationship history fields	Feature
usage_trend_30d, feature_adoption_breadth, seat_utilization	Product-engagement fields	Feature
support_tickets_90d, nps_score	Support & sentiment fields	Feature
treated_discount_offered	Randomized treatment flag	Treatment
renewed	Observed renewal outcome	Target
_true_causal_effect_logodds	Synthetic ground truth (validation only)	Val. only

Cohort at a glance

30,000

total renewal accounts

80.3%

received the discount offer (treated)

74.0%

overall renewal rate

21,000 / 9,000

train / held-out test split

1

randomized binary treatment

Ten features, three business categories

Every input is a field a RevOps or CS platform already tracks — nothing here requires new instrumentation to deploy.



Firmographic

3 features

arr

seat_count

contract_length_months



Relationship history

4 features

tenure_months

prior_discount_count

support_tickets_90d

nps_score



Product engagement

3 features

usage_trend_30d

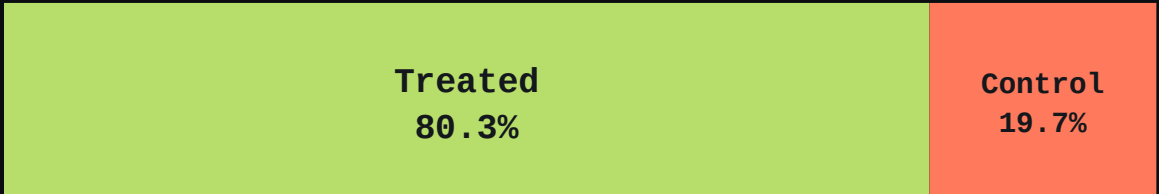
feature_adoption_breadth

seat_utilization

An unequal, but randomized, holdout

80.3% of accounts were randomly offered the discount; 19.7% were randomly held back as a control group. Unequal split sizes are common in real holdout designs — few teams can afford to withhold treatment from half their renewals — and randomization, not the split ratio, is what makes the comparison valid.

Treatment split (n = 30,000)



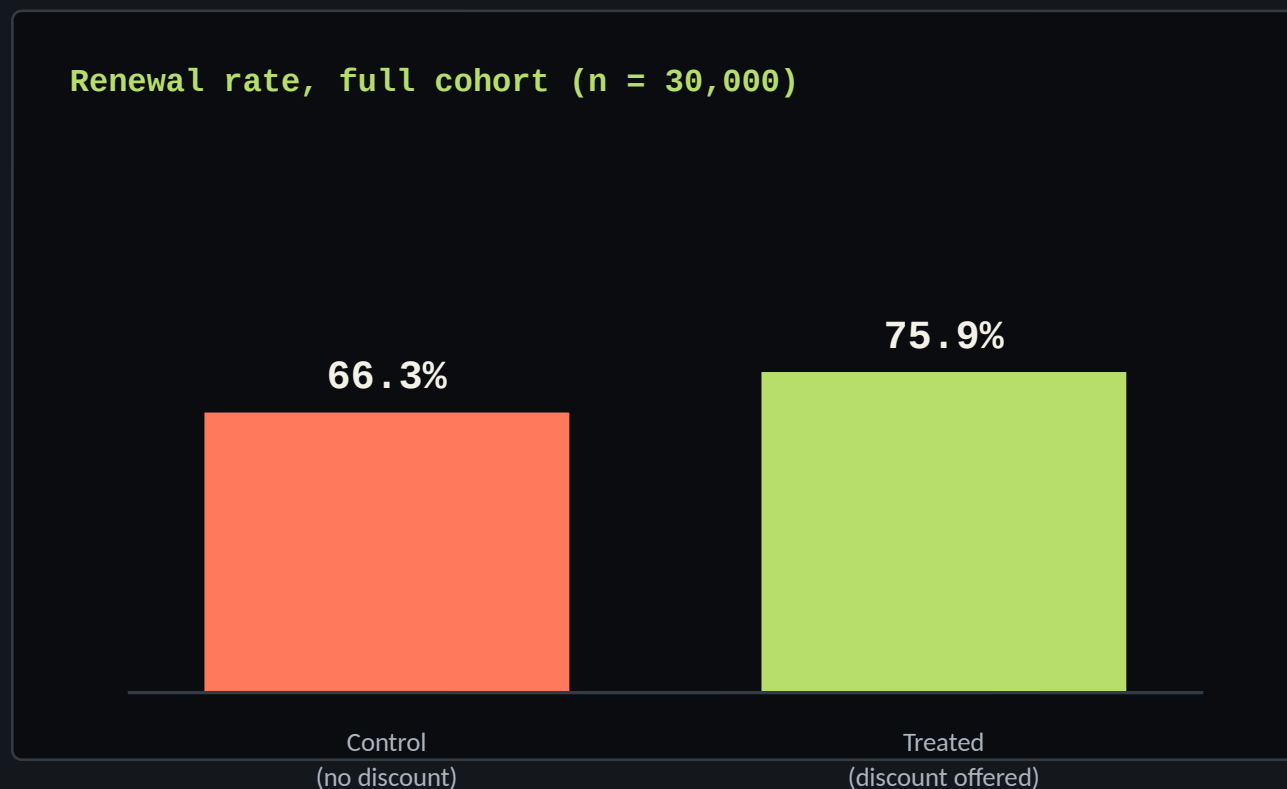
24,079 accounts offered a discount · 5,921 accounts held out as control

Why randomization is non-negotiable

If discounts were historically given to accounts a rep judged "at risk," treated and control accounts would differ systematically — any renewal-rate gap would be confounded, not causal. Randomization is what lets the observed rate difference (Slide 13) be read as a treatment effect at all.

The discount works, on average

Because assignment is random, the raw renewal-rate gap between treated and control is already an unbiased estimate of the average treatment effect — no adjustment needed. The open question isn't whether the discount works on average; it's who it works on.



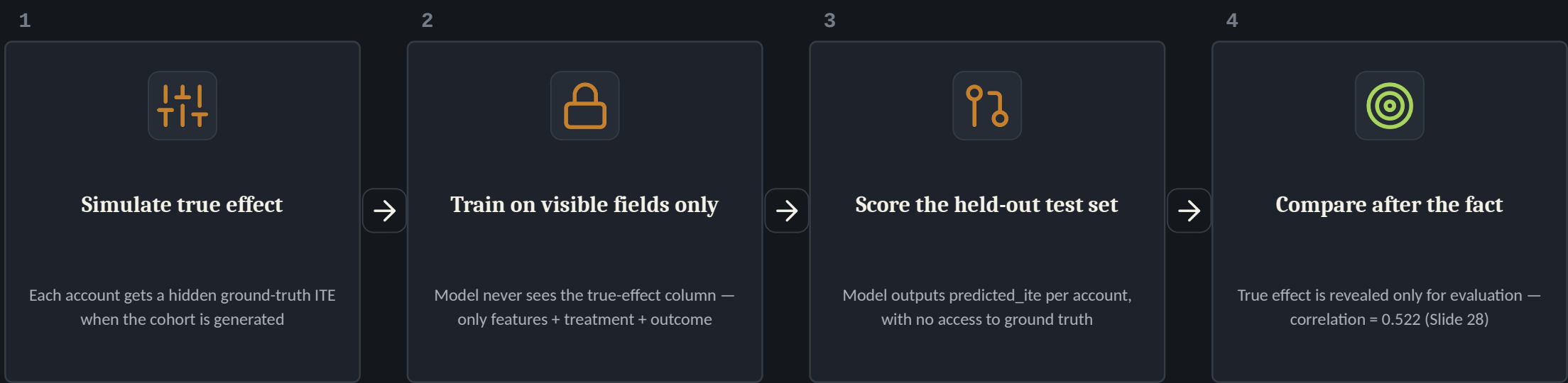
+9.6

points — average treatment effect on renewal rate

This average masks huge heterogeneity — some accounts gain 30+ points, others near zero. The uplift model exists to separate them (Section 04).

A synthetic ground truth, used only to check the model

Every account carries a hidden ``_true_causal_effect_logodds`` value — the actual simulated treatment effect used to generate the data. This column is never available to the model at training or scoring time; it exists solely so this deck can report how well predicted uplift recovers the true effect (Slide 28).



Synthetic, randomized, and disclosed as both



Why synthetic

A generated cohort avoids depending on a real production discount experiment for a portfolio piece, while still requiring genuine causal-inference discipline to build correctly.



Why the design still matters

The randomization, the outcome model, and the evaluation methodology are all real practice — the same pipeline runs unchanged against a real experiment's export.



What the model card says

The model card and EXTENDING.md both state plainly that historical, non-randomized data must never be substituted for this design in production (Slide 39).



A methodological choice, stated up front

Naming "synthetic cohort" and "requires real randomization" as headline caveats — not footnotes — is itself part of demonstrating responsible causal-ML practice.

[03 / 07]

SECTION THREE

Feature Engineering & Segments

Ten fields, three business categories — and one signal that turns out to matter more than any other.

SECTION 3 OF 7



SEC . 03

What each feature category captures



Firmographic

`arr · seat_count · contract_length_months`

Account size and commitment — the features a finance team would reach for first, and the weakest uplift signal in this cohort (Slide 24).



Relationship history

`tenure_months · prior_discount_count · support_tickets_90d · nps_score`

How the account has experienced the relationship so far — including whether it has been discounted before.



Product engagement

`usage_trend_30d · feature_adoption_breadth · seat_utilization`

Recent momentum in how the product is actually used — the single strongest predictor in both response models.

Usage momentum outweighs firmographics

“usage_trend_30d carries roughly 3× the weight of any firmographic field in predicting renewal — whether or not the account was offered a discount.”



Not account size

arr and seat_count rank near the bottom of both response models' feature importance — bigger accounts are not inherently more persuadable.



A recency signal

30-day usage trend outperforms lifetime engagement measures — what an account is doing right now predicts renewal better than its history.

One feature vector, two questions

The same 10-feature vector is asked two separate questions: "how likely to renew if treated?" and "how likely to renew if not?" The gap between the answers is the account's estimated treatment effect — this is the T-learner design detailed in Section 04.



SECTION FOUR

Modeling Approach — T-Learner

Two independent response models and one subtraction — the simplest architecture that estimates heterogeneous treatment effects.



The T-learner: two models, one subtraction

A T-learner trains two entirely independent XGBoost classifiers — one fit only on treated accounts, one fit only on control accounts — then estimates each test account's ITE as the gap between their two predicted renewal probabilities.



Identical configuration, two independent fits

Both response models share one hyperparameter configuration — a shallow, conservative XGBoost setup, slightly gentler than a typical single-outcome classifier to resist overfitting on the smaller control-only training slice.

n_estimators Boosting rounds 250	colsample_bytree Column sampling per tree 0.85
max_depth Shallow trees — controls overfitting 4	objective Calibrated probability output binary:logistic
learning_rate Gentler shrinkage than a single-outcome model 0.06	eval_metric Matches internal validation auc
subsample Row sampling per tree 0.85	random_state Reproducibility 42

Why a T-learner, and where it strains



Simple to reason about

Each response surface is an ordinary binary classifier — no custom loss function, easy to debug and explain to stakeholders.



Clean separation of concerns

The treated and control models can be validated, tuned, and even swapped independently of one another.



A real limitation: regularization bias

When treated (80%) and control (20%) base rates differ this much, each model can systematically over- or under-shrink its predictions.



The disclosed upgrade path

EXTENDING.md proposes replacing this with EconML's CausalForestDML — confidence-interval-aware and less sensitive to the imbalance above.

The two response surfaces value features differently

This divergence is exactly what a T-learner is built to exploit — usage_trend_30d matters roughly 1.5× more for predicting renewal among treated accounts than among control accounts, which is itself a signature of heterogeneous treatment effect.



// WORKED EXAMPLE

Reading a single account's score

p_treated

0.654

Predicted renewal probability if this account is offered the discount

p_control

0.800

Predicted renewal probability if this account is NOT offered the discount

predicted_ite

-0.146

$p_{\text{treated}} - p_{\text{control}}$ — this account is predicted worse off with a discount

This is a real (illustrative) row from the scored test set — a persuadable-looking account by raw churn risk, but the model correctly flags it as already likely to renew, so a discount would be pure margin loss.

SECTION FIVE

Evaluation & Validation

A synthetic ground truth lets this deck do something most uplift deployments never can: check the answer directly.



Three angles on one question: does targeting work?

No single metric captures uplift-model quality. This evaluation combines a ground-truth correlation check (only possible because the cohort is synthetic), the industry-standard Qini curve, and a direct business backtest.



ITE-vs-truth correlation

Directly checks whether predicted uplift tracks the actual simulated effect — a validation that real deployments, lacking ground truth, cannot run.



Qini coefficient

The standard uplift metric: area between the model's targeting curve and random targeting. Threshold-free, rank-based, and comparable across models.



Discount-spend backtest

Translates the ranking into a business number: how much spend is saved by targeting only the top-uplift segment, and how precise that segment is.

Predicted uplift tracks the true effect



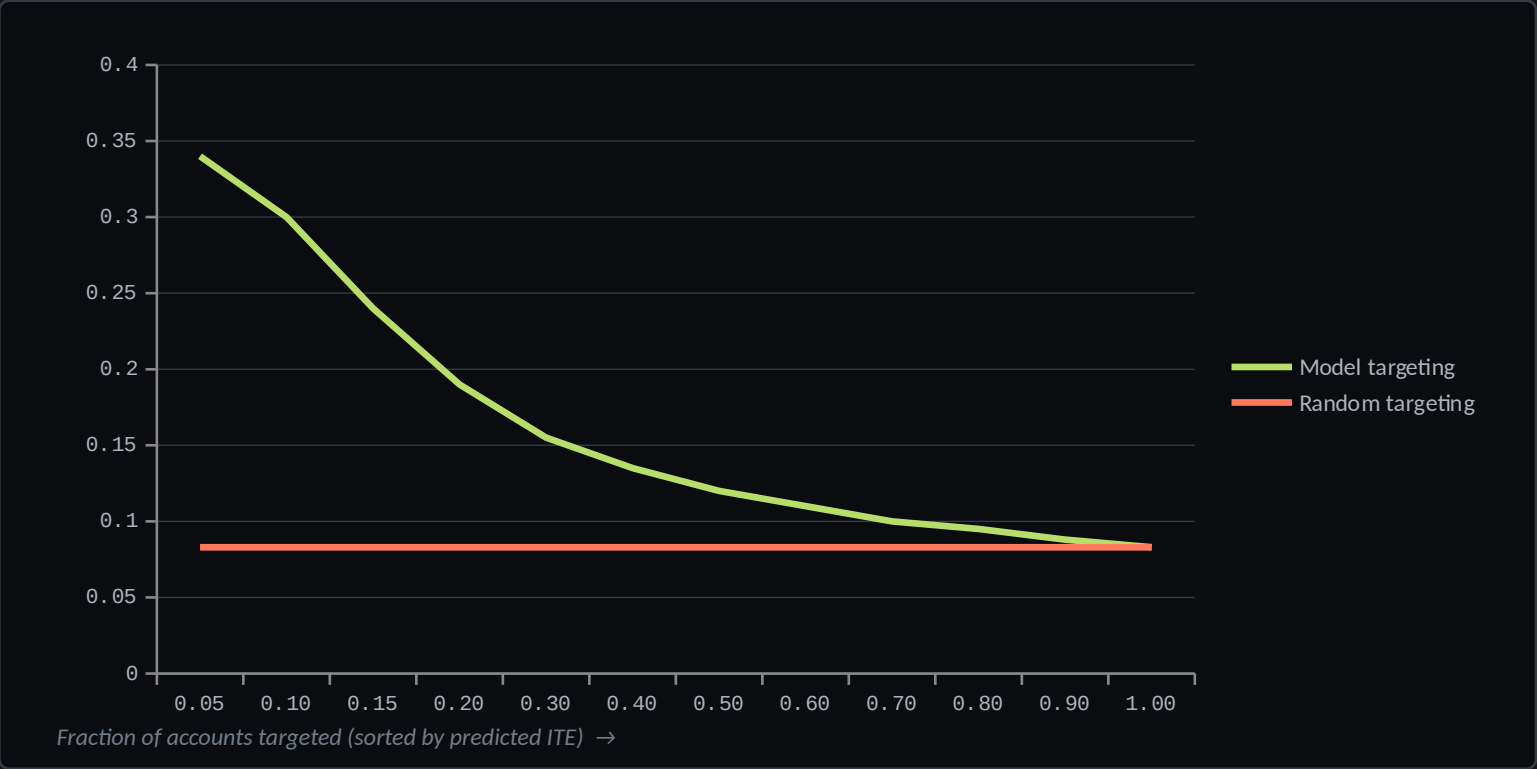
0.522

Pearson correlation, predicted ITE vs. true simulated effect

A moderate, clearly-positive correlation — the model recovers real signal about who benefits, without claiming precision the T-learner architecture can't deliver (see Slide 23's regularization-bias caveat).

The Qini curve

Sort accounts by predicted ITE, then compare the model’s incremental renewal rate at each targeted fraction against random targeting. The gap between the two curves is the Qini coefficient.



0.080

Qini coefficient

Positive and clearly above the random-targeting diagonal — the model’s ranking has real value, even short of the aspirational 0.15 target (Slide 30).

Why 0.080 falls short of the 0.15 target — and what closes it



What the target assumes

The briefing's 0.15 aspirational Qini figure assumes a causal-forest model with confidence-interval-based targeting, not a T-learner (EXTENDING.md, item 1).



Where the T-learner strains

The 80/20 treatment imbalance introduces regularization bias into both response surfaces — exactly the failure mode a causal forest is built to reduce.



The concrete next step

Swap `models_src/t_learner.py` for EconML's `CausalForestDML` — same features, same evaluation pipeline, a structurally better estimator.



Why report a metric that misses its target

A model card that only shows numbers that clear their targets isn't a model card — it's marketing. The gap here is real, disclosed, and has a named fix.

SECTION SIX

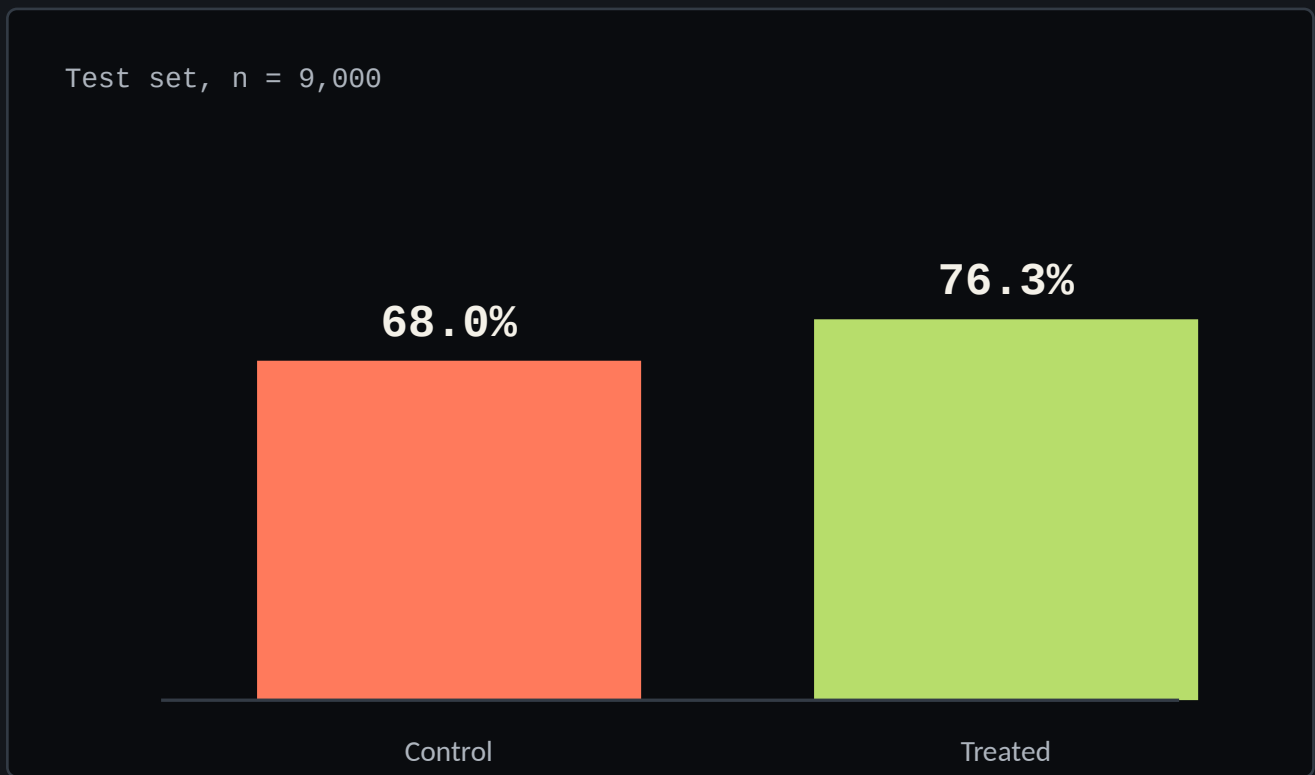
Business Impact & Targeting

Turning a ranking into a spend decision — and checking that the decision actually holds up.



The uplift holds on the test set

The treatment effect seen in the full cohort (Slide 13) replicates cleanly on the 9,000-account held-out test set — the split used for every scoring and evaluation number in this section.



+8.3

points — test-set uplift in renewal rate

Close to the 9.6-point full-cohort estimate — consistent with a clean random split rather than a lucky sample.

// TARGETING RULE

Discount the top 45% by predicted ITE

The threshold is set at the 55th percentile of predicted_ite — targeting the top 45% of accounts by estimated uplift. Everything above the line gets the discount; everything below it doesn't.



45.0%
of test accounts targeted

0.110
predicted ITE cut-off (55th pct.)

Not simply "the biggest accounts"

Targeted accounts renew at a higher rate than the ones skipped — but they carry a lower average ARR, not a higher one. Persuadability, not account size, is driving the targeting rule.

METRIC	TARGETED (TOP 45%)	NOT TARGETED
Renewal rate	76.0%	73.5%
Mean ARR	\$16,509	\$18,873
Segment size	4,050 accounts	4,950 accounts

Less spend, same protective effect, high precision



55 . 0%

Discount-spend reduction vs.
blanket discounting (target $\geq 18\%$)



77 . 5%

Persuadable-segment precision —
true effect > 0.3 log-odds (target $\geq 60\%$)



45 . 0%

Share of accounts still
receiving the discount

What this means in dollars

Skipping the discount on the bottom 55% of accounts by predicted ITE — while 77.5% of the accounts still targeted are confirmed genuinely persuadable — lets a renewal-discount programme protect nearly all of its retention lift at roughly half the spend. The remaining question is how much of that other 22.5% of targeted spend is still justified; a causal-forest upgrade (Slide 30) is the disclosed path to tightening it further.

SECTION SEVEN

Deployment & Responsible Use

A dashboard, a model card, and a hard line on experiment design — causal claims only stand on a real randomized holdout.



The reference dashboard, at a glance

Causal Uplift Renewal-Discount Targeting

Heterogeneous treatment-effect model • T-learner • performance & data overview

REFERENCE IMPLEMENTATION

Recommends renewal discounts by estimated causal effect, not churn propensity

QINI COEFFICIENT

0.080target ≥ 0.15 (naess causal fossat, see EXTENDING.md)

ITE vs TRUE EFFECT

0.522

correlation, synthetic ground truth



SPEND REDUCTION

55%

targeting top 45% by predicted ITE



PERSUADABLE PRECISION

77.5%

targeted segment, true-effect > 0.3



TRAIN / TEST

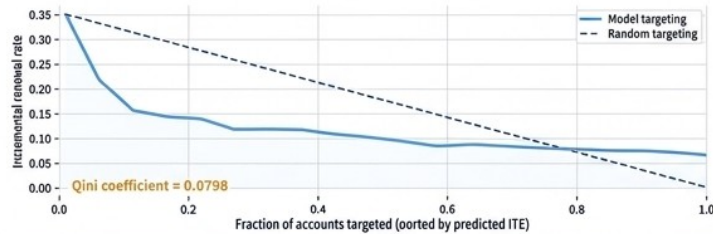
21,000 / 9,000

renewal accounts



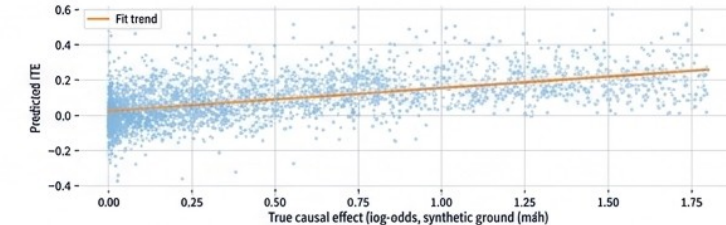
Qini curve

Incremental renewal rate vs. fraction of accounts targeted



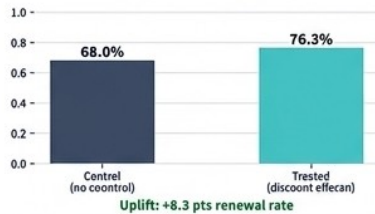
Predicted uplift vs. simulated true effect

n = 9,000 • correlation = 0.522



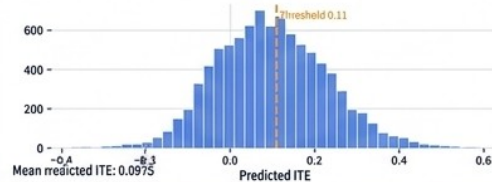
Renewal rate by treatment

Test set — offered discount vs. not



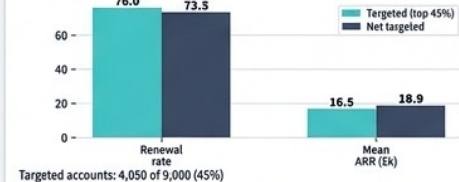
Predicted ITE distribution

Estimated individual treatment effect, test set



Targeted vs. non-targeted segment

Split at top-45% predicted ITE treatment



MODEL CARD SUMMARY

Intended use: recommends which renewing accounts should receive a retention discount, based on estimated causal treatment effect — not churn propensity alone.

Limitations: T-learner can suffer regularization loss when treated control base rates differ; this reg's randomized-holdout design is a real experimental requirement — confounded historical data will NOT give valid valid estimates.

Next steps: swap in EspML CausalForestUML for CI aware targeting (docs/EXTENDING.md), run a real randomized discount offer experiment, extend to multi-tier treatments, wire into CS tooling via webhook.

What it surfaces



Headline KPIs — Qini, ITE correlation, spend reduction, precision



Qini curve vs. random targeting



Predicted ITE vs. simulated true effect



Renewal rate by treatment arm



Targeted vs. non-targeted segment economics

Banner reads: "Recommends which renewing accounts should receive a retention discount — based on estimated causal treatment effect, not just churn propensity."

Model card summary



Intended use

Recommends which renewing accounts should receive a retention discount, based on estimated causal treatment effect — not just churn propensity.



Limitations

T-learner regularization bias under an 80/20 treatment split. The randomized-holdout design is a hard production requirement — confounded historical data will not give valid uplift estimates.



Next steps

Swap in EconML's CausalForestDML, run a real randomized discount-offer experiment, extend to multi-tier treatments, and wire into CS tooling via webhook.

Causal guardrails, not afterthoughts



Randomization is a hard gate

The model card states plainly: confounded historical data will NOT give valid uplift estimates. No production run without a real randomized holdout.



Ground-truth honesty

Every validation number in Section 05 is labeled as relying on a synthetic ground truth unavailable in real deployments — never implied to be a real-world guarantee.



Targets reported even when missed

The Qini coefficient (0.080) is shown against its 0.15 aspirational target and explained, not omitted or reframed as a success.



A named, concrete upgrade path

EXTENDING.md specifies the exact swap — CausalForestDML — rather than a vague "future work" gesture.

// KEY TAKEAWAYS

A disciplined, honest case for causal targeting over blanket discounting



55% less discount spend



77.5% persuadable precision



0.522 ground-truth correlation



Randomization required, no exceptions

Skills demonstrated

Causal inference (T-learner) · Qini / uplift evaluation · randomized-experiment design · honest metric reporting · RevOps-facing dashboarding

Anthony Chinedu Echem

Machine Learning Engineer & Data Scientist



Portfolio contact available on request



THANK YOU